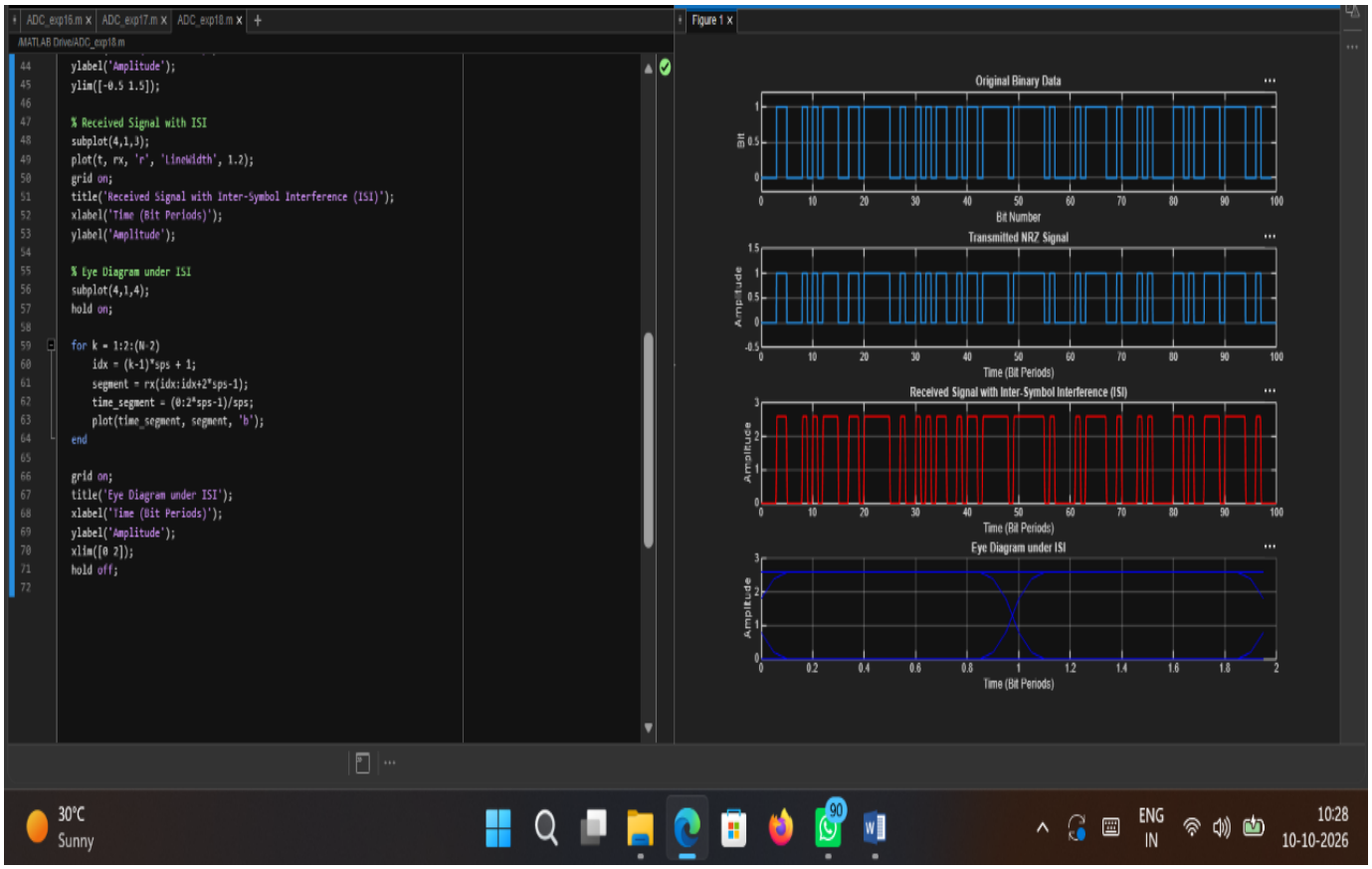
Ex.No:18 - MATLAB SIMULATION AND EVALUATION OF EYE DIAGRAM PARAMETERS UNDER INTER-SYMBOL INTERFERENCE IN DIGITAL COMMUNICATION SYSTEMS

Objective 1:



Code:

```
clc;
clear;
close all;

%% Parameters
N = 100;                    % Number of bits
sps = 20;                   % Samples per symbol

%% Generate Random Binary Data
data = randi([0 1], 1, N);

%% NRZ Encoding
tx = repelem(data, sps);

%% Time Axis
```

```
t = (0:length(tx)-1)/sps;

%% Introduce Inter-Symbol Interference (ISI)
% Simple Low-Pass FIR Channel
h = [0.2 0.6 1 0.6 0.2];

% Received Signal with ISI
rx = conv(tx, h, 'same');

%% Plot Results
figure;

% Original Binary Data
subplot(4,1,1);
stairs(0:N-1, data, 'LineWidth', 1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);

% Transmitted NRZ Signal
subplot(4,1,2);
plot(t, tx, 'LineWidth', 1.5);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
ylim([-0.5 1.5]);

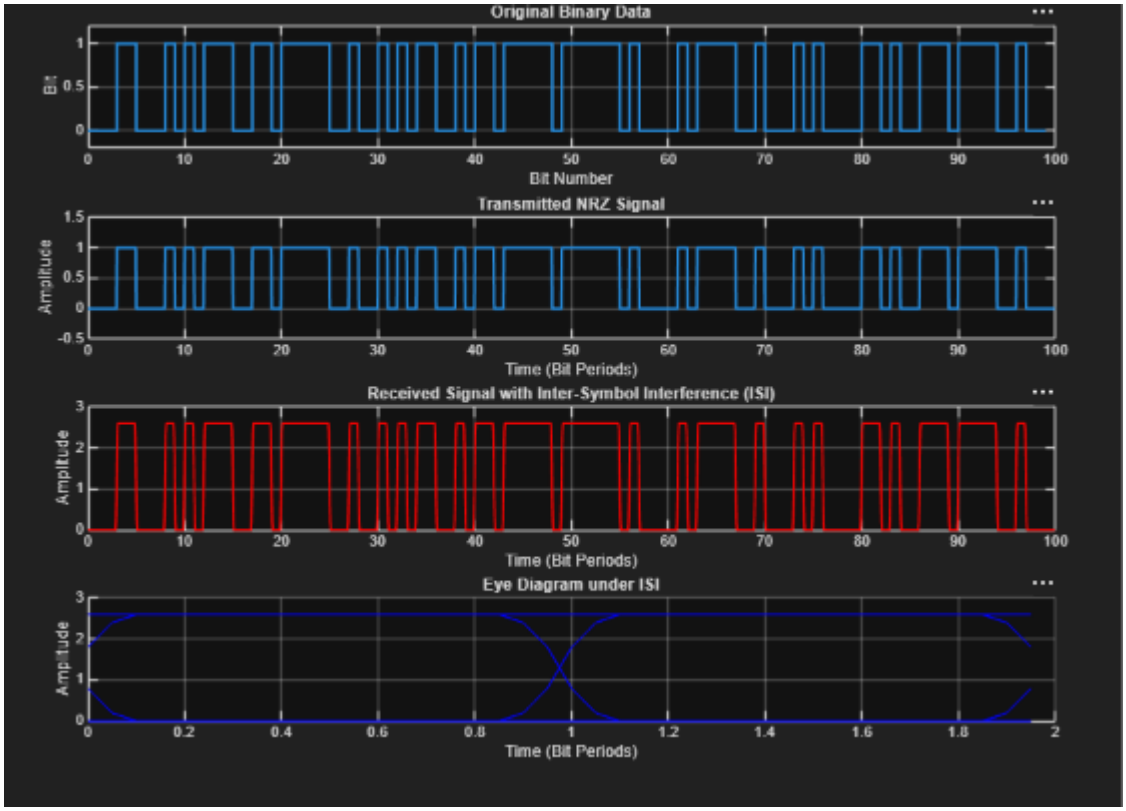
% Received Signal with ISI
subplot(4,1,3);
plot(t, rx, 'r', 'LineWidth', 1.2);
grid on;
title('Received Signal with Inter-Symbol Interference (ISI)');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');

% Eye Diagram under ISI
subplot(4,1,4);
hold on;

for k = 1:2:(N-2)
    idx = (k-1)*sps + 1;
    segment = rx(idx:idx+2*sps-1);
    time_segment = (0:2*sps-1)/sps;
    plot(time_segment, segment, 'b');
end

grid on;
title('Eye Diagram under ISI');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
xlim([0 2]);
hold off;
```

Output:


Original Binary Data
Bit
Bit Number
Transmitted NRZ Signal
Amplitude
Time (Bit Periods)
Received Signal with Inter-Symbol Interference (ISI)
Amplitude
Time (Bit Periods)
Eye Diagram under ISI
Amplitude
Time (Bit Periods)